Mathematical Note — Six-Level Tetrahedral Rhythm (R96, C768, Φ)

Status: Working note (candidate for archival)

Audience: Implementers, auditors, researchers

Keywords: invariant, subgroup action, equivariance, closures, conformance, boundary/bulk, Φ

Abstract

We formalize a minimal mathematical core behind the Six-Level Tetrahedral Rhythm used by Hologram/Φ-Atlas. The note isolates three invariants—**R96**, **C768**, and **Φ-compatibility**—and states clean theorems about (i) a canonical order-2048 boundary subgroup acting on the 12,288-element boundary basis; (ii) closed-form cycle closures for a canonical fair schedule of length 768; (iii) an explicit equivariant tiling of the boundary lattice via a bulk master isomorphism Φ . Deterministic orbit reduction is given as a canonicalization device. We also record conformance-grade statements that match engineering acceptance routines.

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1. Boundary Lattice and Group Action

1.1 Boundary lattice

Let $P = \{0, \ldots, 47\}$, $B = \{0, \ldots, 255\}$, and $G = P \times B$. Then $|G| = 48 \cdot 256 = 12{,}288$.

1.2 Canonical boundary subgroup U_{ref}

Define commuting involutions on G : for pages, $g_{P,0} : p \mapsto p \oplus 1$, $g_{P,1} : p \mapsto p \oplus 2$, $g_{P,2} : p \mapsto p \oplus 4$; for bytes, $g_{B,k} : b \mapsto b \oplus 2^k$ for $k = 0, \ldots, 7$. These maps preserve $0 \le p < 48$ and $0 \le b < 256$. Let

$$U_{\mathrm{ref}} = \langle g_{P,0}, g_{P,1}, g_{P,2}, g_{B,0}, \ldots, g_{B,7} \rangle \cong (\mathbb{Z}/2\mathbb{Z})^{11}.$$

Proposition 1 (Order). $|U_{\mathrm{ref}}| = 2^{11} = 2048$.

Proof. The generators commute and are independent involutions; the map from $\{0,1\}^{11}$ to U_{ref} taking a bit string to the corresponding word is an isomorphism of groups. □

Proposition 2 (Free action on orbits used below). For the six base points $S = \{(0,0), (8,0), (16,0), (24,0), (32,0), (40,0)\} \subset G$, the stabilizer in U_{ref} is trivial; hence each orbit has size 2048.

Proof. The only word sending $(8m, 0)$ to itself must zero all byte XOR bits and page low XOR bits; commutativity and independence force the identity word. □

Corollary 3 (Six-orbit tiling). The six orbits of the points in S are pairwise disjoint and cover G. In particular, $6 \cdot 2048 = 12{,}288 = |G|$.

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2. Canonical Fair Schedule and Closures (C768)

2.1 Canonical fair schedule

Define $\sigma_{\mathrm{CFS}} : \{0, \dots, 767\} \to G$ by $\sigma_{\mathrm{CFS}}(t) = (t \bmod 48,\ t \bmod 256)$. Then each page appears exactly 16 times and each byte exactly 3 times. Moreover, for any prefix of length a multiple of 48 (resp. 256), page (resp. byte) counts differ by at most 1.

2.2 Closure family

For observables of the form $f(p, b) = d + \alpha_p + \gamma_b$ with $d \in \mathbb{Z}$, $\alpha \in \mathbb{Z}^{48}$, $\gamma \in \mathbb{Z}^{256}$, define

$$S_r = \sum_{t=0}^{767} t^r\, f(\sigma_{\mathrm{CFS}}(t)) \quad (r = 0, 1,\ 2).$$

Theorem 4 (Closed forms for CFS-768). With $N = 768$,

$$S_0 = 768\, d + 16 \textstyle\sum_p \alpha_p + 3 \sum_b \gamma_b,$$

$$S_1 = \tfrac{(N-1)N}{2}\, d + \textstyle\sum_p \alpha_p (16p + 5760) + \sum_b \gamma_b (3b + 768),$$

$$S_2 = \tfrac{(N-1)N(2N-1)}{6}\, d + \textstyle\sum_p \alpha_p (16p^2 + 11520p + 2{,}856{,}960) + \sum_b \gamma_b (3b^2 + 1536b + 327{,}680).$$

Proof sketch. Partition the index set by congruence classes modulo 48 and 256, use arithmetic-progression identities for $\sum t$ and $\sum t^2$, and the exact hit counts 16 and 3. All coefficients follow by direct algebra. □

Remark. For the constant observable $f \equiv 1$, one recovers the exact integers $S_0 = 768$, $S_1 = 294{,}528$, $S_2 = 150{,}700{,}160$.

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3. Equivariant Φ-Compatibility

Let a bulk space be $A \times \mathbb{Z}_2^{10}$ where $A = \{0, 1, 2, 3, 4, 5\}$ indexes six page buckets and the 10-bit fiber controls two page XORs and 8 byte XORs. Define $\Phi : A \times \mathbb{Z}_2^{10} \to G$ by

$$\Phi(a, \mathbf{z}) = (\, p\, ,\, b\,) \quad \text{with} \quad p = 8a \oplus z_0 \oplus (z_1 \ll 1) \oplus (z_* \ll 2), \quad b = \sum_{k=0}^{7} (z_{2+k} \ll k),$$

where $z_* \in \{0,1\}$ is a dedicated bulk bit for $g_{P,2}$. Let U_{ref} act on the bulk by flipping the corresponding fiber bit.

Theorem 5 (Equivariance). For each generator $u \in U_{\text{ref}}$ and bulk state x , $\Phi(u \cdot x) = u \cdot \Phi(x)$.

Proof. By construction: toggling a fiber bit in bulk applies the corresponding XOR to the boundary coordinates. ☐

Corollary 6 (Boundary-first tiling). The six U_{ref} -orbits of the images of $(a, \mathbf{0})$ for $a = 0, \ldots, 5$ are disjoint and cover G . Hence Φ is boundary-first and satisfies tiling.

Label preservation. Any label function on G that is U_{ref} -invariant is automatically preserved along bulk–boundary transport via Φ .

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4. Deterministic Orbit Reduction

Fix an ordered generator list for U_{ref} and a cryptographic hash H (e.g., SHA-256) on byte encodings of representatives. Let $\mathcal{G} = \{0,1\}^{11}$ enumerate group elements via Gray code so successive masks differ by one bit.

Definition 7 (Canonical representative). For any representative x , define

$$\operatorname{canon}(x) = \operatorname*{argmin}_{g \in \mathcal{G}} H(g \cdot x)$$

with ties broken by first occurrence along the Gray traversal.

Proposition 8 (Determinism & uniqueness). Given a fixed generator order, fixed hash, and fixed byte encoding, $\operatorname{canon}(x)$ is deterministic and unique.

Proof. Gray code visits each g exactly once; H induces a total order on byte strings, so a unique minimum exists. The traversal and hash are fixed, hence the output is deterministic. ☐

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5. Resonance Classes (R96)

Let a **selector** be a length-8 toggle vector with a pinned bit and polarity. A **normalization** operator N (pair-normalized unity) maps selectors to canonical representatives and commutes with the boundary subgroup. Let the class map be

$$\rho(s) = \operatorname{Orbit}_U(N(s)) \in \mathcal{R}.$$

Specification Theorem 9 (R96 invariants, implementation-level). For admissible selectors S of size 256, any normalization N that (i) is involutive on canonical reps, (ii) commutes with U , and any deterministic orbit reducer as in §4, yields

$$|\mathcal{R}| = 96, \qquad \text{and multiplicities } \{2:32,\ 3:64\}.$$

Status. This is treated as a **conformance invariant**: implementations must pass enumeration under the acceptance routine; a structural proof reduces to analyzing selector stabilizers under U and the normalization equivalences. (A complete proof is deferred.)

Invariance under unity choice. Replacing the unity quartet by an isomorphic quartet preserves the cardinality and multiplicity split, because the change is mediated by a relabeling automorphism commuting with U .

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6. Conformance Statements (Mathematical Form)

C768. An implementation passes **C768** iff, for σ_{CFS} and a generating set of observables containing 1 , all page indicators, and all byte indicators, the empirical (S_0, S_1, S_2) match Theorem 4's closed forms exactly (integer equality) or within machine-epsilon-scaled tolerances for floating accumulators.

Φ-compatibility. An implementation passes **Φ** iff there exists Φ and generators of U s.t. equivariance holds on a spanning set and the induced orbits of a complete set of Φ -images partition G .

R96. An implementation passes **R96** iff deterministic enumeration yields exactly 96 classes with multiplicity multiset $\{2^{\times 32}, 3^{\times 64}\}$ and the result is invariant under isomorphic unity quartets.

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7. Complexity, Determinism, and Auditing

- **Complexity.** Enumeration for R96 is linear in $|S|$ with orbit reduction cost $O(|U|)$ per class search (here $|U| = 2048$). C768 is $O(768 \cdot |\mathcal{F}|)$. Φ-compatibility is dominated by orbit partitioning; Schreier–Sims–style caching reduces repeated evaluations.
- **Determinism.** All routines are deterministic given the configuration bundle: unity anchors, generator order, hash salt, and labeling seed.
- **Audit bundle.** A machine-readable bundle records configuration, schedule digest, generator certificate (order 2048), closure values, Φ commutation and tiling logs, and R96 enumeration outcomes.

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8. Proof Sketches and Identities (Collected)

- **AP sums:** $\sum_{t=0}^{N-1} t = N(N-1)/2$, $\sum_{t=0}^{N-1} t^2 = (N-1)N(2N-1)/6$ with $N = 768$ give $294{,}528$ and $150{,}700{,}160$.

- **Page residues:** For fixed p , the page times are $p + 48k$ for $k = 0, \dots, 15$; summing yields coefficients 16 , $16p + 5760$, and $16p^2 + 11520p + 2{,}856{,}960$ in Theorem 4.
- **Byte residues:** For fixed b , the byte times are $b + 256q$ for $q = 0, 1, 2$; summing yields coefficients 3 , $3b + 768$, and $3b^2 + 1536b + 327{,}680$.
- **XOR closure for pages:** On $0..47$ the high bits 16 and 32 never co-occur, so flipping low bits 1,2,4 preserves range.

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9. Open Problems and Directions

1. **Structural proof of R96.** Give a group-theoretic derivation of the 64/32 multiplicity split from selector stabilizers and normalization relations.
2. **Beyond affine observables.** Characterize the maximal class of observables for which closed forms exist under σ_{CFS} .
3. **Alternative subgroups.** Classify all order-2048 subgroups of $U(48) \times U(256)$ that commute with the same Φ and compare tilings.
4. **Bulk geometry.** Relate the six buckets $A = \{0, \dots, 5\}$ to positive-geometry cells and study spectral implications for resonance labeling.

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10. Minimal Pseudocode (Deterministic Canonicalization)

```
# Gray-code orbit reduction (canonical representative)
def orbit_reduce(rep_bytes, apply_gen):
    best = (sha256(rep_bytes).digest(), rep_bytes)
    prev = 0
    cur = rep_bytes
    for mask in gray_code(11):
        diff = prev ^ mask
        if diff:
            idx = (diff.bit_length() - 1)
            cur = apply_gen(idx, cur)
            h = sha256(cur).digest()
            if h < best[0]:
                best = (h, cur)
        prev = mask
    return best[1]
```

Acknowledgments

This note distills the spec-grade annex into a compact mathematical core suitable for certification proofs and interoperability.